Chipsee

Industrial PC

PPC-A53-121



PN: CS10768-IMX8MP-121P

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PPC-A53-121



Front View



Rear View



Side View 1



Side View 2

1 Product Overview

The Cortex®-A53 series PPC-A53-121 (PN: CS10768-IMX8MP-121P) is a high-quality industrial panel PC. This single board computer features a 12.1" 10-point capacitive touch screen with a resolution of 1024 x 768 pixels and a brightness of 350 cd/m^2.

Key Applications

- Human Machine Interface HMI
- Mobile Applications
- Video Processing
- Machine Learning
- Video Gaming
- Process Control
- Process Monitoring
- ATM...

It is available as a device hosed in an aluminum casing with bezels.

The PPC-A53-121 Industrial Panel PC is based around the powerful i.MX8MP System on Chip (SoC), powered by the NXP i.MX8MP low-power processor which integrates a quad-core Cortex®-A53 1.6GHz processor.

The i.MX8MP supports multi-format video decoders and has a high-performance LPDDR4 4GB RAM capable of sustaining demanding memory bandwidths. It also provides a complete set of peripheral interfaces.

2 Ordering Options

Chipsee products can be customized during the ordering process. The product will be shipped with the pre-installed factory defaults if no extra requirements are specified. The table in the Hardware Features section provides information about the default options bundled with the product.

> **Note**
>
> You can order the PPC-A53-121 from the official Chipsee Store or from your nearest distributor.

2.1 Operating System

This product comes with a pre-installed OS of your choice. Please see the list below for the supported OSes, which can also be obtained from the Software Documentation section, along with the detailed installation instructions.

- Android 12
- Yocto Linux Qt 6.3

> **Warning**
>
> The Software Documentation section provides a detailed instruction on how to install different OSes on your own. However, bear in mind that Chipsee can't take the responsibility of inadequate installation procedure. If you "brick" your device, please contact Chipsee Technical Support at **support@chipsee.com** for further assistance.

2.2 Optional Features

Does not include the 4G/LTE module by default. The module is optional and can be selected at the Chipsee store during the ordering process.

> **Warning**
>
> Installation, repair, and maintenance tasks should be performed by trained personnel only. Chipsee does not bear any responsibility for damage caused by inadequate handling of the product.

3 Hardware Features

The PPC-A53-121 Industrial Panel PC offers a broad range of performance and connectivity options for scalable integration, providing expandability according to future needs. Some of the key features are listed in the table below.

Key Features

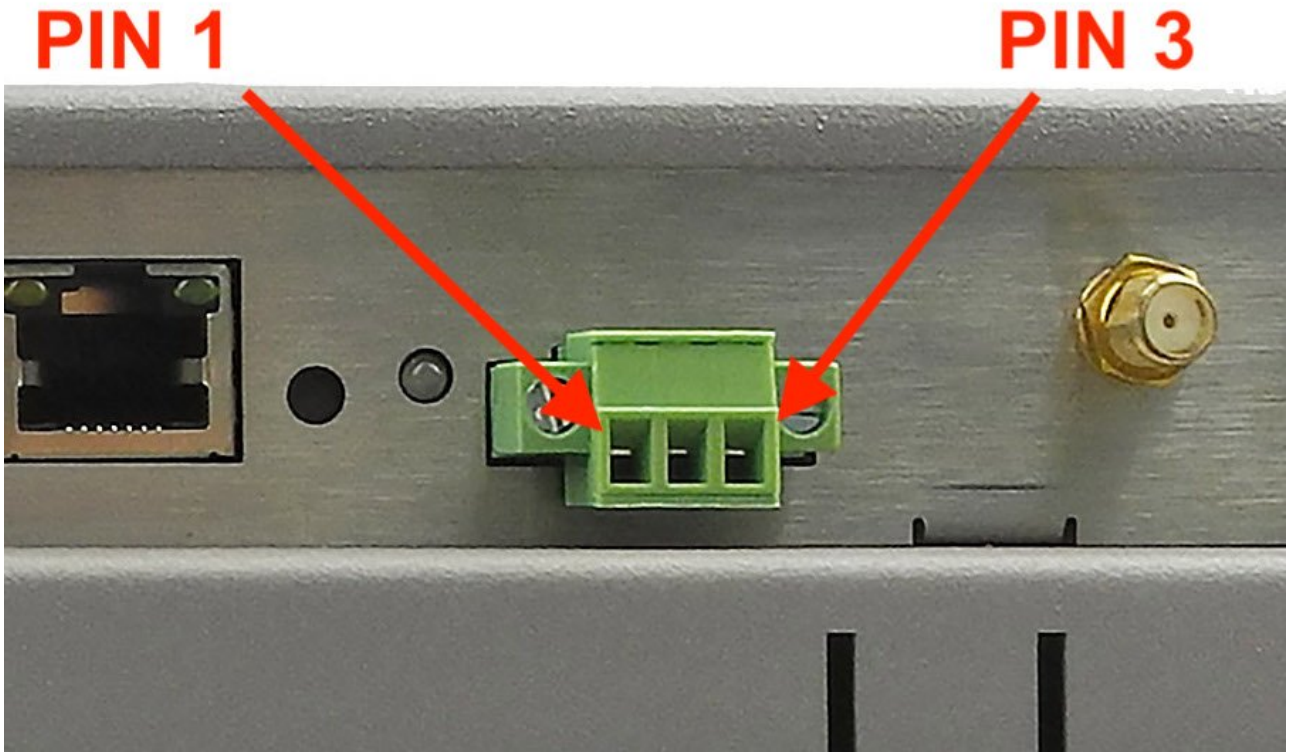
PPC-A53-121	
CPU	NXP i.MX8MP, Quad(4)-core Cortex-A53 (1.6GHz)
RAM	LPDDR4 4GB
eMMC	32GB
SSD	Not supported
Storage	TF Card, Supports up to 128GB SDHC
HDMI	1 x HDMI OUT
Display	12.1" LCD, 1024 x 768, High Brightness: 350 cd/m^2
Touch	10-point capacitive touch screen
USB	1 x USB 2.0 HOST, 1 x USB 3.0 HOST, 1 x USB Type-C[1]
LAN	1 x RJ45, GbE
Audio	3.5mm Audio In/Out Connector, 2W Internal Speaker
Buzzer	Yes
RTC	High accuracy RTC with farad capacitor, can work 1 week after power off **(default)**. High accuracy RTC with lithium coin battery, can work 3 years after power off **(optional)**.
RS232	Default 2 x RS232 **(Optional 6 x RS232 at most, include 1 debug port)**[2]
RS485	Default to 2 x RS485. Optionally, these 2 x RS485 can be configured to RS232.[2]
CAN	Default to 2 x CAN. Optionally, these 2 x CAN can be configured to RS232.
GPIO	8 Channels Isolated IO, 4 x Input and 4 x Output
WiFi/BT	Integrated WiFi/BT Module (Bluetooth is functional on Yocto Linux, not functional on Android)
4G/LTE	Supported, Optional
Power Input	From 15V to 36V
Current	0.80A Max at 15V
Power Consumption	12W Typical.
Working Temperature	From -20°C to +70°C
OS	Android 12, Yocto Linux Qt 6.3
Dimensions	320 * 260 * 55mm
Weight	3100g
Mounting	Panel, VESA

[1] USB3.0 port and USB-C port share one node and cannot be used together. In Linux USB3.0 HOST is enabled by default; in Android USB-C OTG is enabled by default. You can change this config by software in the operating systems, e.g.: disable USB3.0 HOST then enable USB-C OTG in Linux, or vice versa in Android.

[2] (1,2)
This product has 3 x CPU UART, 1 x USB UART by default, and 6 x UART terminals (RS232/RS485) at most. The default configuration is 2 x RS232 and 2 x RS485, including 1 RS232 debug port. This product has 2 x CAN by default, all of them can be configured to RS232 (USB UART). UART can be swapped between RS232 and RS485 modes easily, if you need a different RS232/RS485/CAN configuration, please get in touch with the Chipsee Technical Support at **support@chipsee.com** when placing an order.

4 Power Input

The PPC-A53-121 Industrial Panel PC can be powered by a wide range of input voltages: From 15V to 36V DC. The power input connector is a **3-pin, 3.81mm terminal**. The polarity and the pinout is shown in the figure below.



Power Input

Note that the "**+**" sign represents the positive power input, it is printed at the casing and as a silk-screen on a PCB of the embedded version. The "**-**" terminal is shorted to the ground.

Power Connector

Power Input Definition		
Pin Number	**Definition**	**Description**
Pin 1	Positive Input	DC Power **Positive Terminal**
Pin 2	Negative Input	DC Power **Negative Terminal**
Pin 3	Ground	**Power System Ground**

> **Note**
>
> The system ground "**G**" is connected to power negative "**-**" on board.

5 Touch Screen

The PPC-A53-121 Industrial Panel PC uses a 10-point capacitive touch screen. The touch layer is connected through USB.

> **Attention**
>
> A capacitive touch screen is susceptible to power noise and Electromagnetic Radiation (EMR). It may cause LCD ripples or even capacitive touch malfunction. If using a capacitive multi-touch test application, you might notice the touch points float erratically across the display. There are several solutions to this problem:
>
> 1. Use a high-quality Power Adapter Unit (PSU) with low EMR. You can also provide power from a battery.
> 2. Make sure that the PPC-A53-121 Power Input connector (pin 3) is properly connected to the Power System Ground to provide sufficient EMI shielding and eliminate the problem entirely.
> 3. Bad GND problems can also be confirmed by touching pin 3 of the Power Input connector with one hand while operating the capacitive touch screen with the other hand. In this case, the operator's body acts as the Power System Ground.

6 Connectivity

There are many connectivity options available on the PPC-A53-121 industrial PC. It has 1 x USB 2.0 HOST, 1 x USB 3.0 HOST, 1 x USB Type-C (USB3.0 and USB-C share one node); 1 x RJ45, GbE; up to 6 x UART terminals (RS232/RS485), up to 2 x CAN.

6.1 RS232/RS485/CAN

The serial communication interfaces (RS485, RS232, and CAN) are routed to a **24-pin 3.81mm terminal**, as illustrated on the figure below.

2	4	6	8	10	12	14	16	18	20	22	24
CAN1_H	CAN1_L	IN4	IN3	IN2	IN1	OUT4	OUT3	OUT2	OUT1	GND_ISO	VDD_ISO
1	3	5	7	9	11	13	15	17	19	21	23
CAN0_H	CAN0_L	RS485_4-	RS485_4+	RS485_3-	RS485_3+	RS232_0_RXD	RS232_0_TXD	RS232_2_RXD	RS232_2_TXD	GND	VCC5V



RS232, RS485 and CAN

This product has 3 x CPU UART, 1 x USB UART by default, and 6 x UART terminals (RS232/RS485) at most. The default configuration is 2 x RS232 and 2 x RS485, including 1 RS232 debug port. This product has 2 x CAN by default, all of them can be configured to RS232 (USB UART).

Connectivity Section

RS232 / RS485 / CAN / GPIO Pin Definition:		
Pin Number	**Definition**	**Description**

RS232 / RS485 / CAN / GPIO Pin Definition:		
Pin 1	CAN0_H	CAN H signal
Pin 2	CAN1_H	CAN H signal
Pin 3	CAN0_L	CAN L signal
Pin 4	CAN1_L	CAN L signal
Pin 5	RS485_4-	USB UART1, RS485 -(B) signal
Pin 6	IN4	Isolated Input 4
Pin 7	RS485_4+	USB UART1, RS485 +(A) signal
Pin 8	IN3	Isolated Input 3
Pin 9	RS485_3-	CPU UART3, RS485 -(B) signal
Pin 10	IN2	Isolated Input 2
Pin 11	RS485_3+	CPU UART3, RS485 +(A) signal
Pin 12	IN1	Isolated Input 1
Pin 13	RS232_0_RXD	CPU UART4, RS232 RXD signal
Pin 14	OUT4	Isolated Output 4
Pin 15	RS232_0_TXD	CPU UART4, RS232 TXD signal
Pin 16	OUT3	Isolated Output 3
Pin 17	RS232_2_RXD	CPU UART2, Debug RXD signal
Pin 18	OUT2	Isolated Output 2
Pin 19	RS232_2_TXD	CPU UART2, Debug TXD signal
Pin 20	OUT1	Isolated Output 1
Pin 21	GND	System Ground
Pin 22	GND_ISO	Isolated Ground
Pin 23	VCC5V	System 5V output, up to 1A
Pin 24	VDD_ISO	Isolated Power Input, supports 5V~24V

Attention

1. This product supports changing 2 x RS485 to 2 x RS232, supports changing 2 x CAN to 2 x RS232, providing up to 6 x RS232 (including one debug port).
2. The GPIO has been optical isolated and it uses the 24V logic by default. You can use an external isolated power input but the power input range should be from 5V to 24V DC.
3. The 4 GPIO output channels can drive at most 500mA current on each channel.

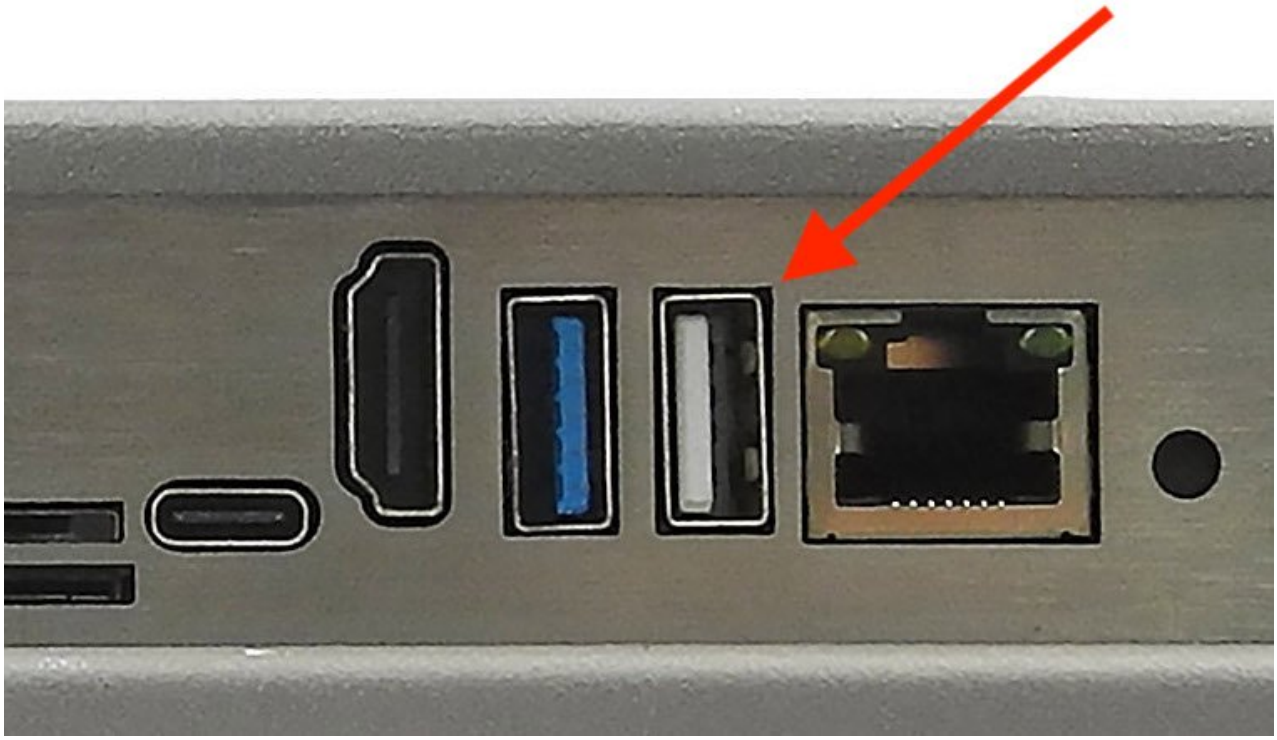
Attention

1. RS485 can control the input and output direction automatically. There's no need to control it from within the software.
2. The 120Ω match resistor for the **RS485** is **already mounted** by default.
3. The 120Ω match resistor for the **CAN** bus is **NOT mounted** by default.

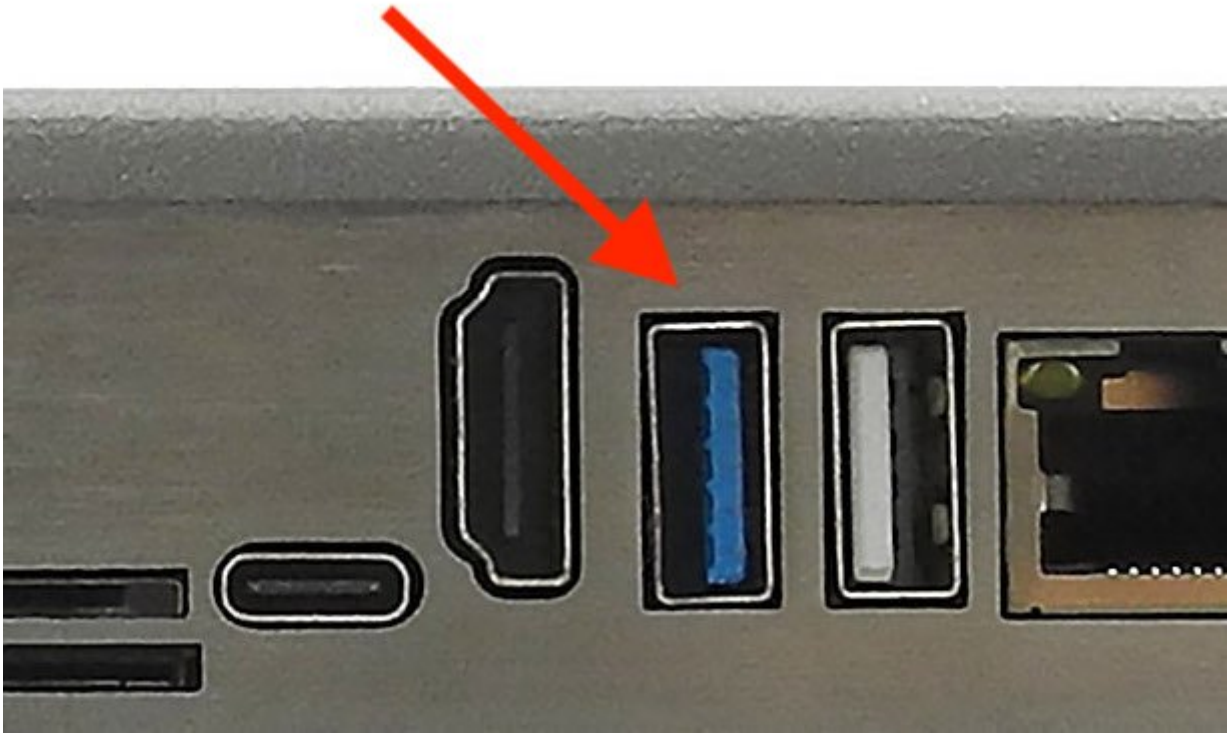
6.2 USB Connectors

There are 2 x **USB HOST** and 1 x **USB DEVICE** (for flashing OS) ports onboard: 1 x USB 2.0 HOST, 1 x USB 3.0 HOST, 1 x USB Type-C, as shown in the figures below.

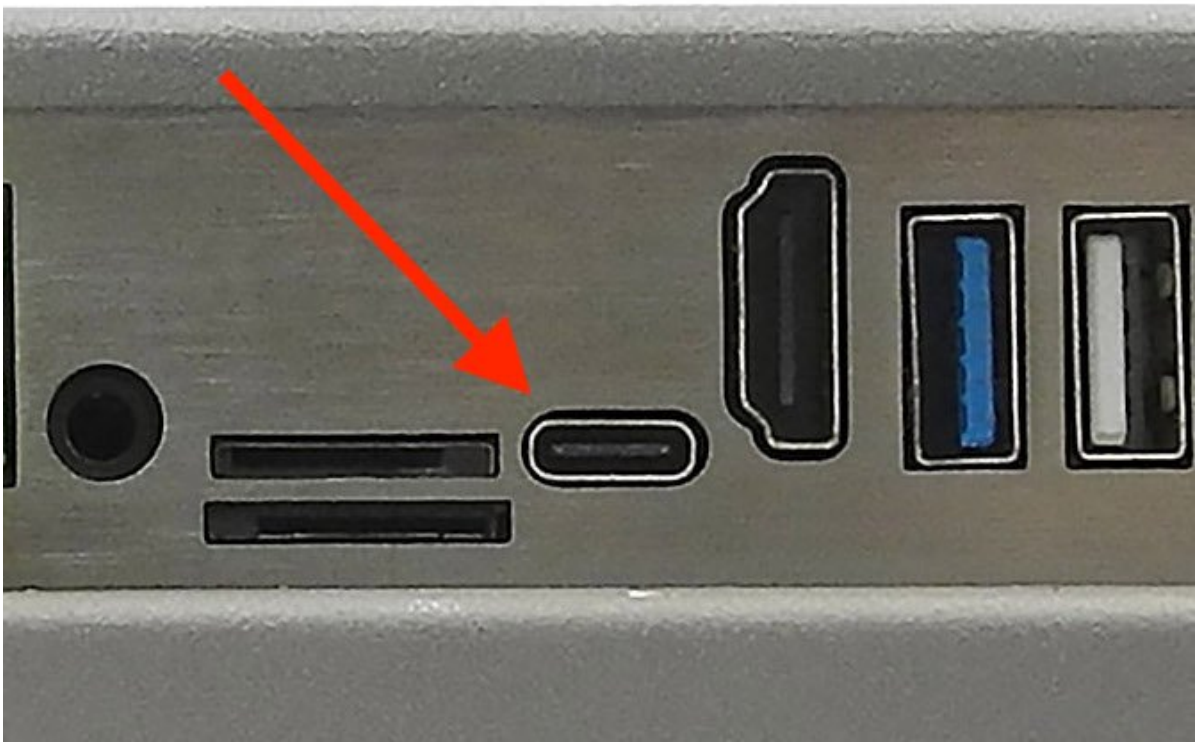
USB3.0 port and USB-C port share one node and cannot be used together. In Linux USB3.0 HOST is enabled by default; in Android USB-C OTG is enabled by default. You can change this config by software in the operating systems, e.g.: disable USB3.0 HOST then enable USB-C OTG in Linux, or vice versa in Android.



USB 2.0 HOST Port



USB 3.0 HOST Port



USB Type-C Port

Warning

Be careful not to touch surrounding electronic components accidentally while plugging USB devices into the embedded IPC version.

6.3 LAN Connectors

LAN (RJ45) connector provides Ethernet connectivity over standardized Ethernet cables as shown in the figure below. The integrated 1 x RJ45, GbE interface supports up to 1 Gbps data throughput.

This product doesn't support PoE.



RJ45 LAN Connector

> **Note**
>
> Use CAT5 or better cables to achieve full data throughput over maximum distance defined by the 1000BASE-T standard (100m).

6.4 WiFi & BT Module

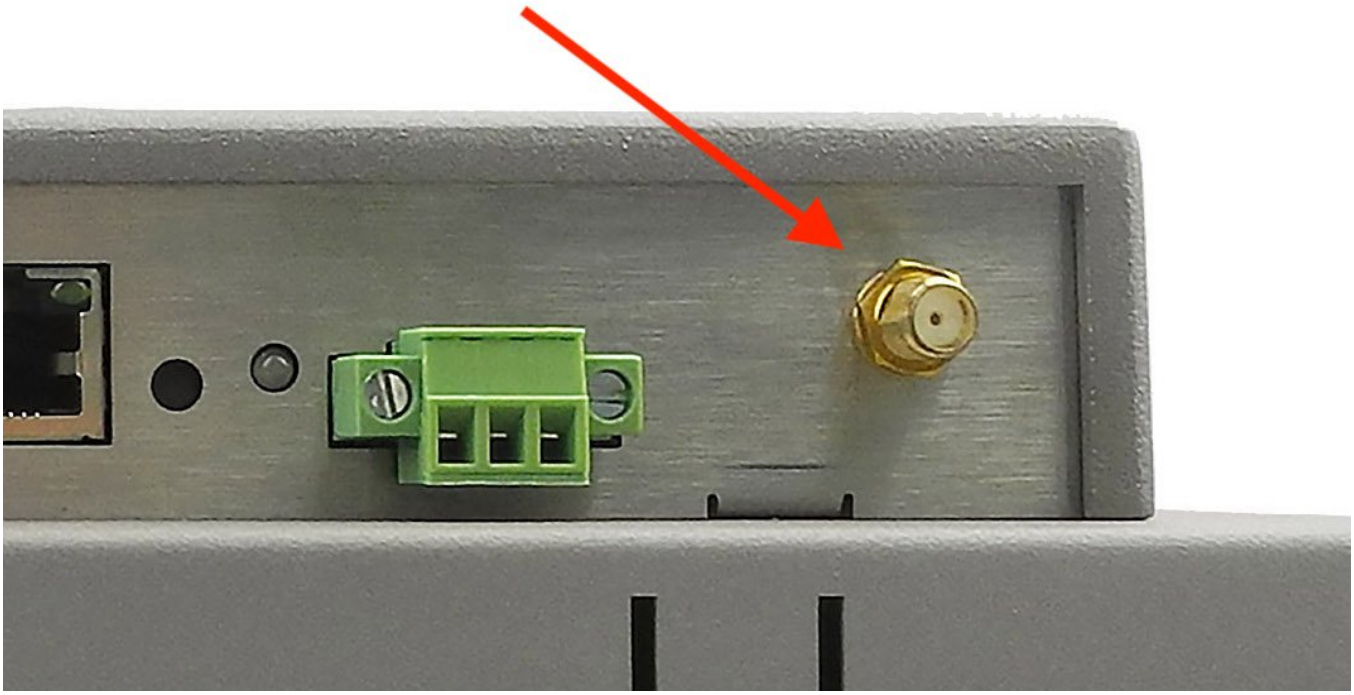
The PPC-A53-121 Industrial Panel PC is equipped with the popular **Realtek RTL8821CS WiFi/BT module** which supports BT/BLE 2.1/3.0/4.2, as well as 802.11ac/abgn 433Mbps 2.4/5.8 GHz Wireless LAN (WLAN).

> **Warning**
>
> Bluetooth is functional on Yocto Linux, **not** functional on Android OS.



RTL8821CS WiFi/BT Module

The PPC-A53-121 includes an SMA connector for an external WiFi/BT antenna, as illustrated in the figure below.



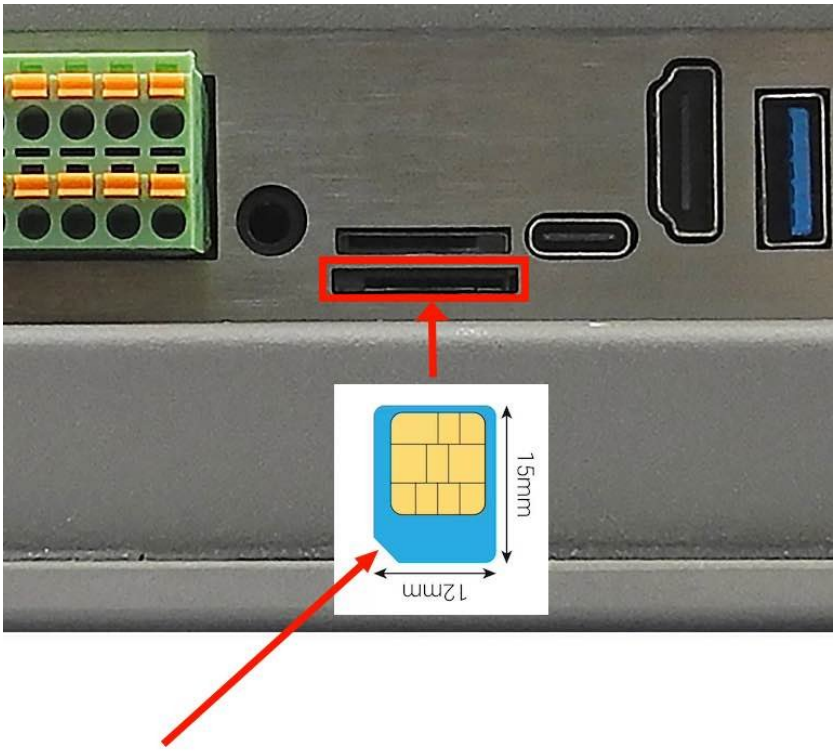
WiFi+BT Antenna SMA

6.5 4G/LTE Module

The PPC-A53-121 Industrial Panel PC is equipped with a **mini-PCIe connector** that can connect a 4G/LTE module. The customer will also need a SIM Card Holder and a 4G/LTE Antenna Connector to ensure 4G/LTE works on the PPC-A53-121. SIM card does **NOT** support hot plug. **Power off** before inserting or removing SIM card.



mini-PCIe Connector & 4G/LTE Module



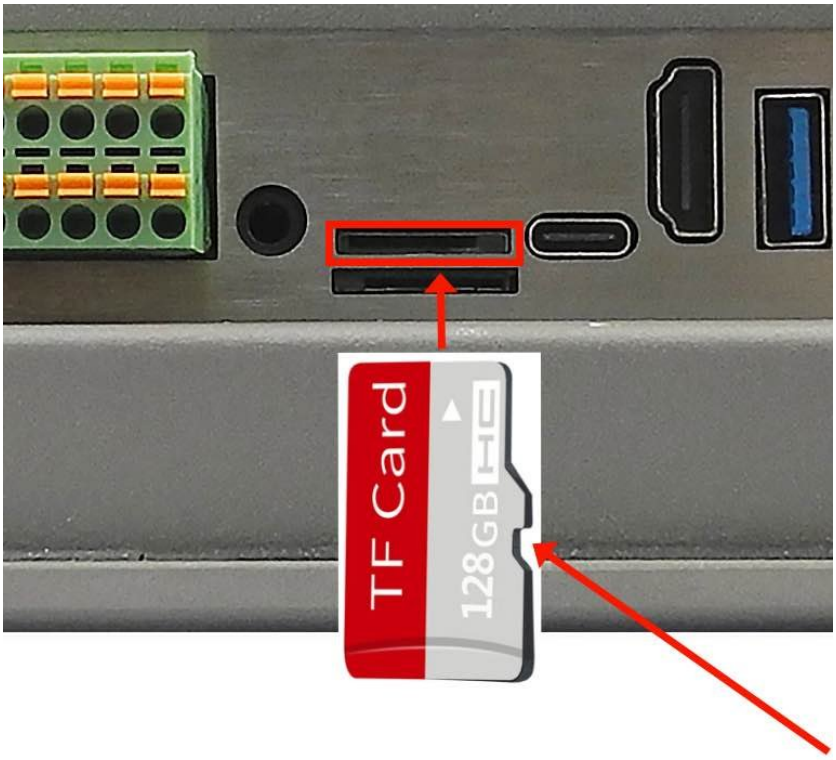
SIM Card Direction (Micro SIM Card)

Attention

The product does not come shipped with the 4G/LTE module by default. The customer can choose the 4G/LTE module option when placing an order, we will install all the necessary components.

7 TF Card Slot

The PPC-A53-121 Industrial Panel PC features 1 x **TF Card (micro SD) slot**. TF Card can address up to 128GB of storage.

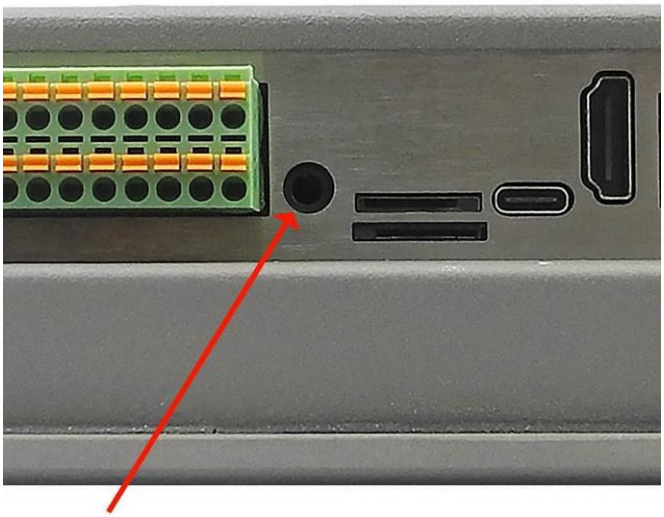


TF (micro SD) Card Slot

> **Note**
>
> The product does not come shipped with the TF Card by default.

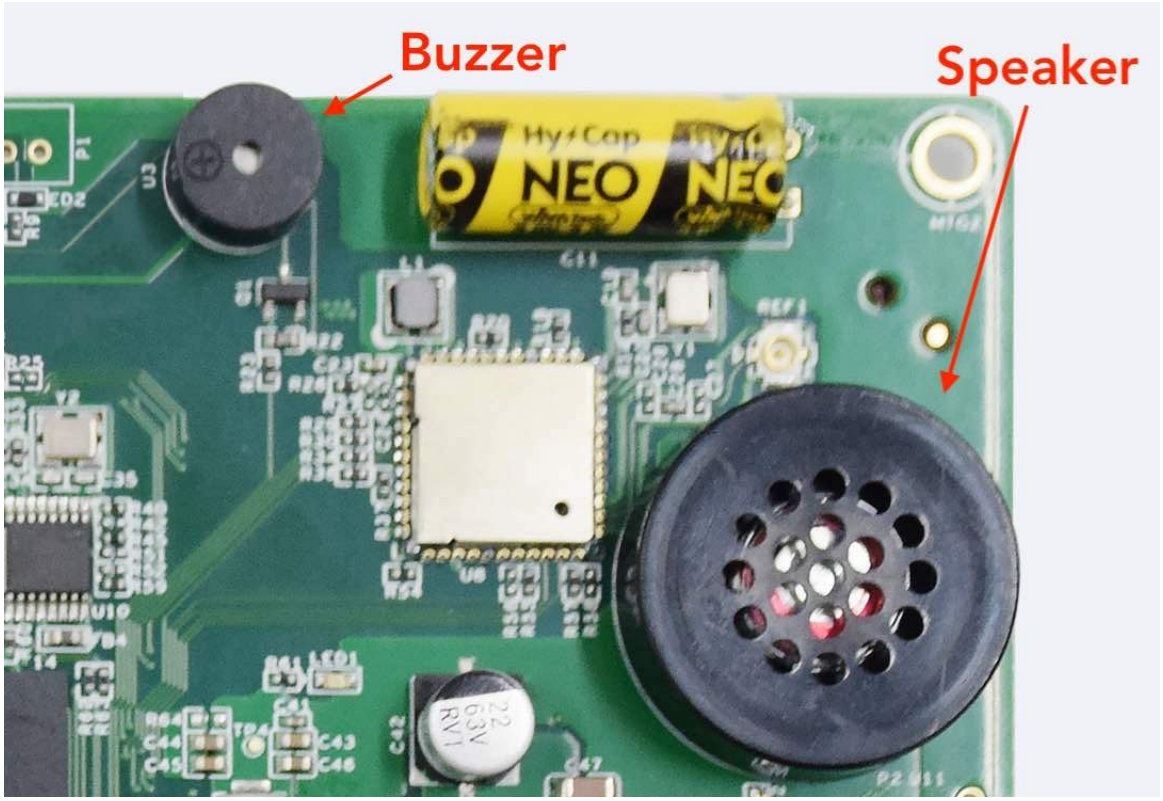
8 Audio Connectors

The PPC-A53-121 Industrial Panel PC features some audio peripherals. It has a **3.5mm audio input/output jack**, an **internal speaker**, as well as a small **buzzer**.



Audio Connector

The miniature 2W embedded speaker is handy for audio reproduction, the small buzzer can play alarm/ notification sounds.



2W Micro Speaker and Buzzer

> **Attention**
>
> By plugging in the headphone cable, the internal speaker will be disabled automatically.

9 HDMI Connector

The PPC-A53-121 Industrial Panel PC is equipped with 1 x HDMI OUT connector. The HDMI connector allows connecting an additional (external) monitor. HDMI output resolution can be configured by the software.



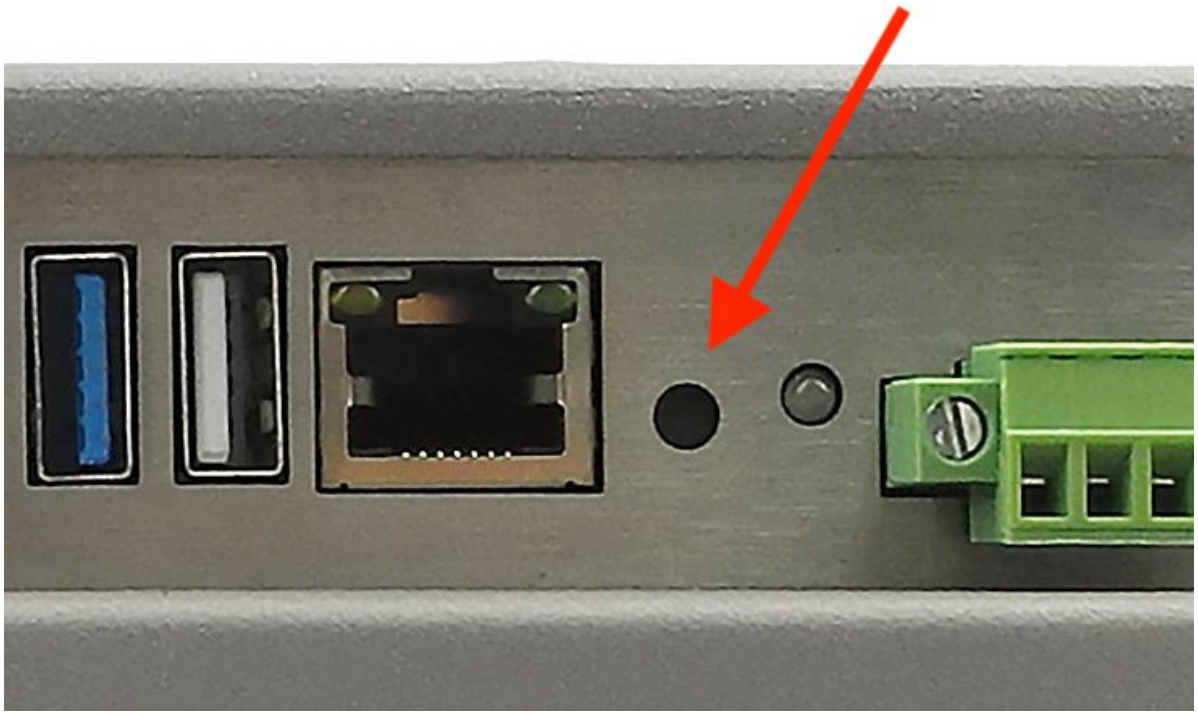
HDMI Connector

10 PROG Button

The PPC-A53-121 Industrial Panel PC has one button on the board marked as PROG, as shown in the figure below. It controls how the device will be booted.

To boot from SD card, press and hold the PROG button, then connect the power supply, after a few seconds, you can see the system boot from SD card, then you may release the button.

When the button is not pressed while powering up, the PPC-A53-121 will boot normally from eMMC.



PROG Button

11 Mounting Procedure

You can mount PPC-A53-121 with VESA mounting (guide): **100 x 100** mm, 4 x **M4** (6mm) screws, enabling simplified installation onto any standard mounting fixture.

You can also mount PPC-A53-121 with panel mounting method (guide).

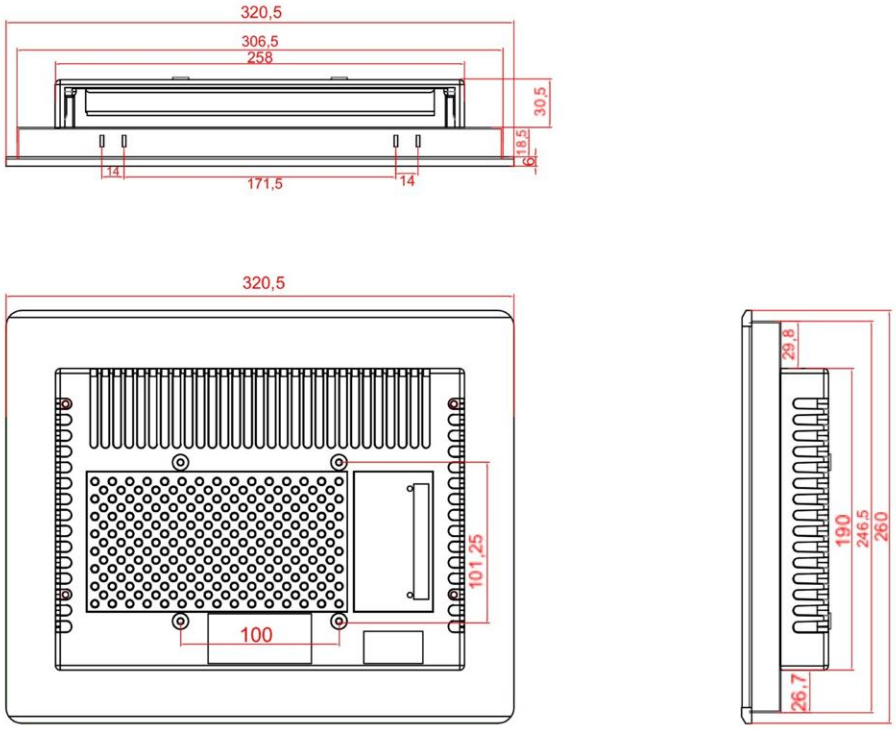
> **Attention**
>
> Please make sure the display is not exposed to high pressure when mounting into an enclosure.

12 Mechanical Specifications

For PPC-A53-121, the outer mechanical dimensions are 320 * 260 * 55mm (W x L x H).

The cutout dimensions of this 12.1 inch product is 311mm x 251mm.

Please refer to the technical drawing in the figure below for details related to the specific product measurements.



PPC-A53-121 *Technical Drawing*

13 Disclaimer

This document is provided strictly for informational purposes. Its contents are subject to change without notice. Chipsee assumes no responsibility for any errors that may occur in this document. Furthermore, Chipsee reserves the right to alter the hardware, software, and/or specifications set forth herein at any time without prior notice and undertakes no obligation to update the information contained in this document.

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14 Technical Support

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